

Benjamin Johnston

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EXPERIENCE

Software Engineer, Self-Employed

Jul 2021 – Present

- Author and maintain 10+ public systems-level libraries and tools spanning multi-architecture assemblers (x86, x86_64, MIPS, CIL), JIT machine-code generation, and reverse-engineering research across C, C++, C#, and D.
- Reverse-engineered Windows syscalls and undocumented embedded firmware APIs using static disassembly (IDA Pro, Ghidra), manual export-table walking, and dynamic shellcode generation to recover internal behavior without vendor documentation.

RPA Engineer & Model Evaluator, Handshake · Contract

Nov 2025 – Jun 2026

- Built AutoHandshake!, a Selenium-based automation suite that eliminated 95% of manual annotation and triage effort across 400+ contractor hours through unattended execution, failure recovery, and audit trails.
- Mentored contractors on platform operations, rubric interpretation, and efficient evaluation workflows.

Expert Contributor, Snorkel AI · Contract

Dec 2025 – May 2026

- Engineered constraint-based training workflows across 2 major contracted projects, training Claude and ChatGPT models on complex programming and logic tasks.
- Executed comprehensive CI/CD testing and validation protocols via Docker containers, ensuring robust deployment stability and training set quality.
- Designed rigorous evaluation frameworks to systematically assess model reasoning capabilities, driving continuous improvements in code generation and logical deduction accuracy.

OPEN SOURCE PROJECTS

VEXAPI — Reverse Engineer & Author | C++

2024, [GitHub](#)

- Reverse-engineered the VEX Robotics V5 firmware's undocumented private API via ARM64 disassembly of the VEXos system image, recovering jump tables and driver signatures for display, sensor, and motor control, enabling low-level robot programming beyond the official SDK.

Multidasm — Author & Maintainer | D

2026, [GitHub](#)

- Built a compile-time assembler in D targeting CIL, x86, x86_64, and MIPS instruction sets, exposing a fluent, strongly typed instruction API with label support and machine-code emission; AMD64 backend spans MMX, SSE, AVX, and AVX2.

UndeadInterop — Author & Maintainer | C#

2026, [GitHub](#)

- Built a Windows syscall research library using manual export-table walking and dynamic shellcode generation to invoke native APIs without static imports, including hook detection across user-mode and kernel-mode syscall variants.

Jitsort — Author | C

2025, [GitHub](#)

- Designed a JIT sorting algorithm in C that emits customized x86_64 assembly directly into executable memory, using array values as memory indexes for constant-time sorting; minimal, dependency-free C API supports custom element sizes.

TECHNICAL SKILLS

Languages: C, C++, D, C#, x86/x86_64 & ARM64 Assembly, MIPS Assembly, Python, JavaScript, SQL

Tools & Frameworks: IDA Pro, Ghidra, Git, CMake, Docker, CI/CD, Linux

Concepts: Reverse engineering, firmware & embedded API analysis, binary disassembly, syscall/API hooking, JIT code generation, multi-ISA assembler design, performance optimization

Practices: Agile, Kanban, Test-Driven Development, Code Review, Open Source Contribution

EDUCATION & CERTIFICATIONS

Columbus State Community College — AAS, Software Development

Expected Dec 2026

Certifications: Handshake Model Validation 2 — Expert; GitHub Foundations (PMI); Microsoft Career Essentials (Generative AI, Software Development, Azure AI, Security); SQL Programming